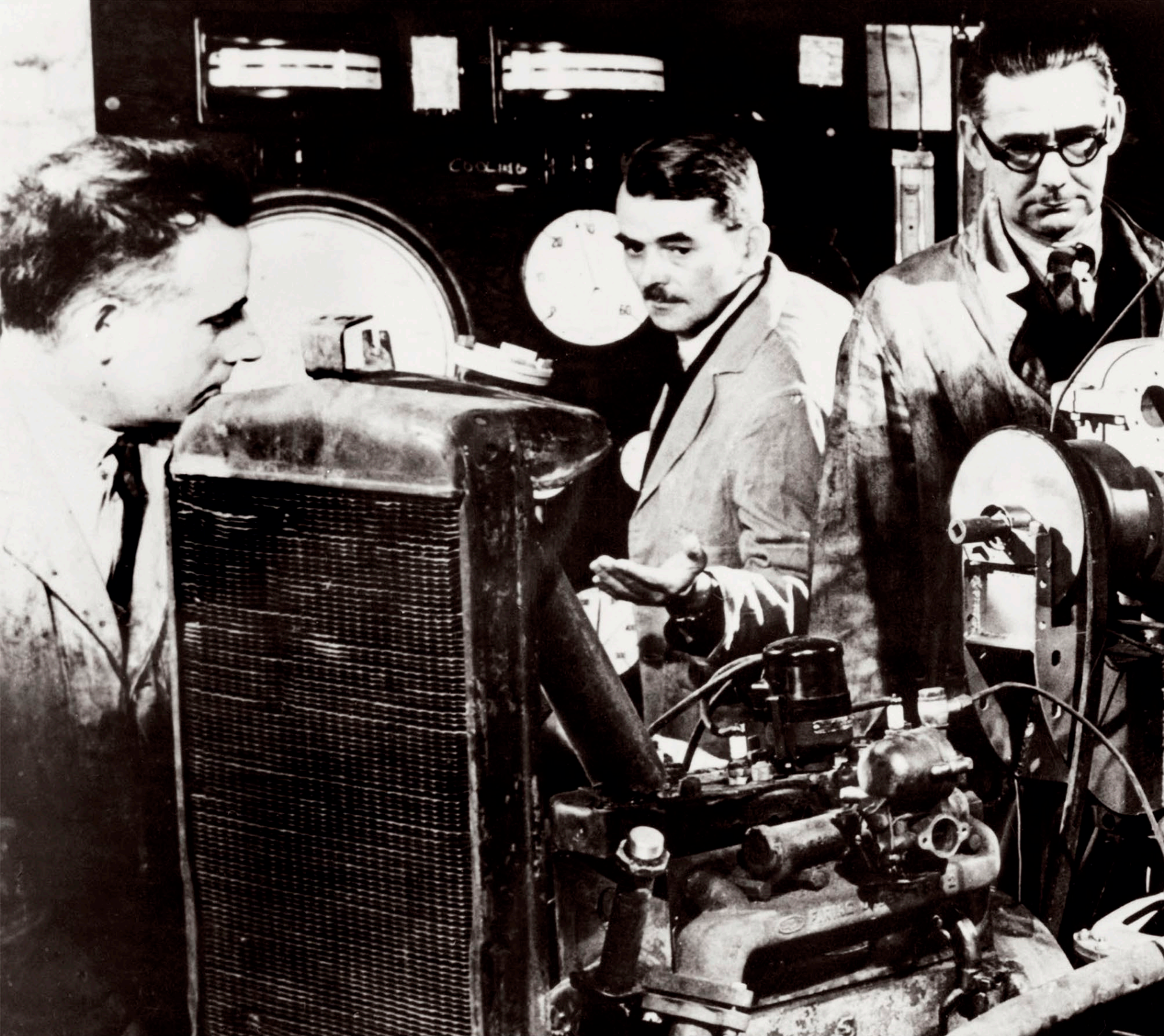




Frank Whittle's jet engine

British Air Commodore Sir Frank Whittle's invention—the jet engine—transformed air travel, allowing millions to do what had seemed unthinkable a few decades before: to cross the Atlantic at speed. In 1920, when Whittle presented his engine design to the Air Ministry, they rejected it. Despite this setback, Whittle still patented his “turbojet engine” in 1930. Whittle's engine used a turbo-driven compressor wheel to force air into a combustion chamber, and relied on the exhaust jet efflux for thrust. In 1936 Whittle secured financial backing and, with the Air Ministry's approval, formed Power Jets Ltd in Lutterworth, Leicestershire. Meanwhile, another engineer in Germany, Hans von Ohain, had independently devised a jet engine that flew in 1939, though he

patented his plans after Whittle. The two men are now given joint recognition of the invention. The first successful flight of Whittle's engine took place on 15 May 1941 in an aircraft that had been specifically designed for the purpose—the experimental Gloster E28/39. This paved the way for the Gloster Meteor, the first production jet, powered by the Power Jets W2 engine. After World War II, the turbojet engine was applied to passenger planes, allowing faster journeys in larger aircraft. Boeing led the field, with its Model 707 jetliner entering service in 1958.

Frank Whittle (center) tests an engine in 1946 with colleagues G. B. Bozzoni and H. Harvard in the documentary film *Jet Propulsion*.